SAR Drone Search Parameter Optimisation Decision Flow

AENGM0074 Group Design Project

University of Bristol — 2025–26

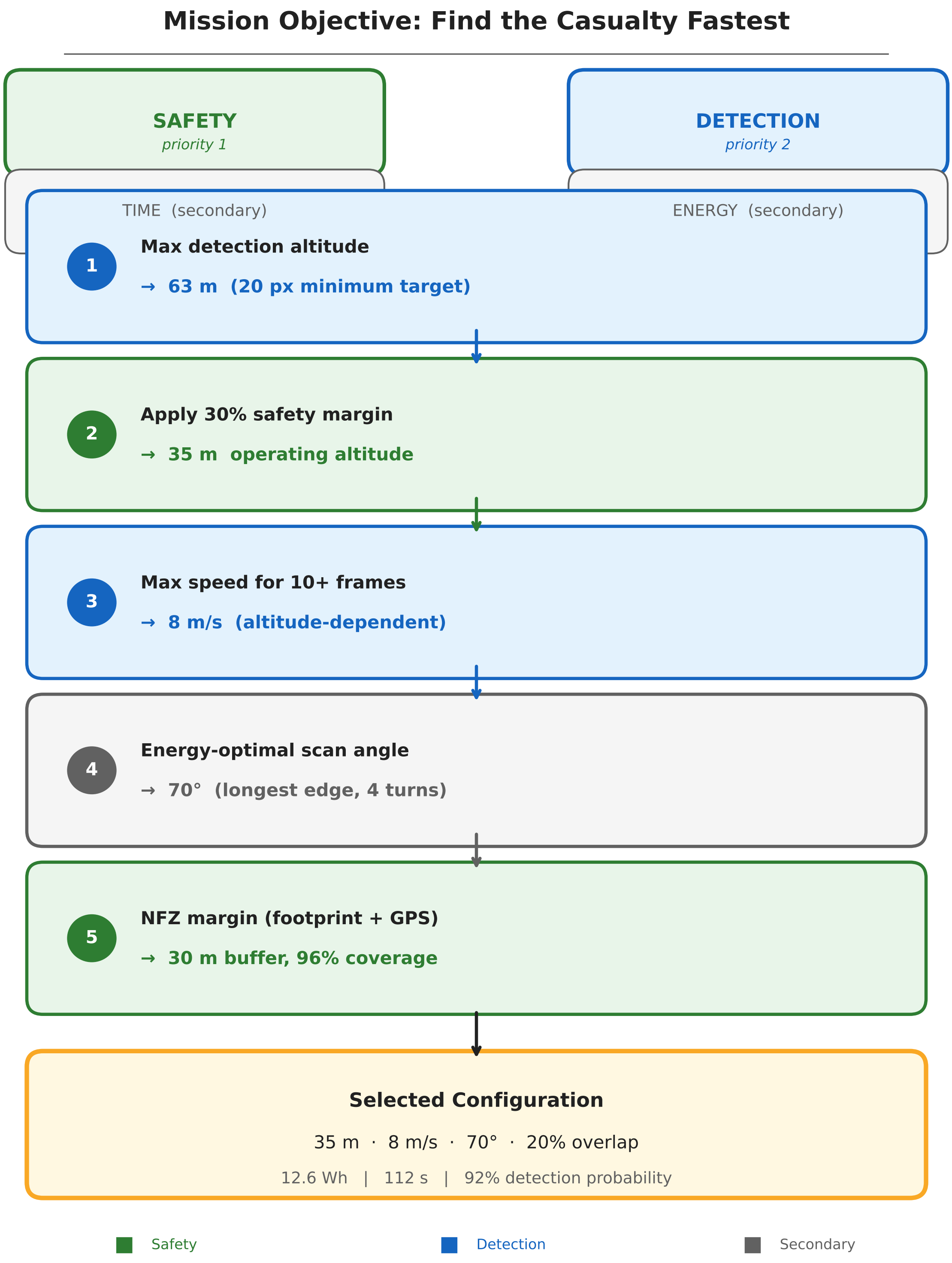

Figure 1: Decision flow infographic: the sequential chain from target size through altitude, speed, scan angle, and NFZ margins to the final operating configuration.

1 The Problem

A hiker is missing. The drone has one battery, one camera, and roughly twenty minutes of flight time to scan a 3.2-hectare field bordered by a protected wildlife reserve. Five things compete:

- **Coverage** — scan every square metre of the search area.
- **Detection** — actually see a person-sized target from altitude.
- **Time** — the "golden hour" of hypothermia survivability is ticking.
- **Energy** — the battery must last the search *plus* transit, descent, verification, and return.
- **Safety** — the drone must never enter the adjacent SSSI no-fly zone.

No single configuration maximises all five simultaneously. Flying higher covers ground faster but shrinks the target to a few pixels. Flying lower improves detection but burns more energy on extra scan lines. Flying near the boundary maximises coverage but risks an incursion.

We adopted a clear priority stack: **safety** > **detection** > **speed** > **energy**. An NFZ violation grounds the project. A missed casualty defeats its purpose. A slow search wastes the golden hour. High energy use is undesirable but survivable.

2 The Variables We Control

Five parameters define the search pattern. Each affects multiple objectives; altitude alone touches all five.

Table 1: Decision variables and the objectives they influence.

Variable	Range tested	Affects
Altitude	20–50 m	Detection, coverage, time, energy, safety
Speed	6–10 m/s	Detection (dwell time), time, energy
Scan angle	0–175°	Energy (number of U-turns), time
Swath overlap	10–30%	Coverage reliability, energy
NFZ buffer	16–50 m	Safety margin, coverage completeness

The coupling structure is shown in Figure 2: darker cells indicate stronger influence. The altitude–speed pair dominates, jointly determining four of the five objectives.

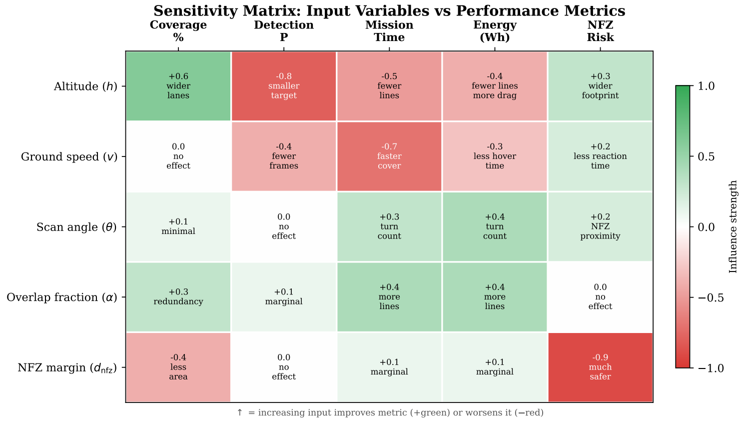

Figure 2: Sensitivity matrix: darker cells indicate stronger influence of each input parameter on each derived metric.

3 Our Strategy: Safety-First, Detection-Optimised

Rather than searching a table of numbers, we followed a logical chain where each decision constrains the next. The reasoning runs in six steps:

1. Start from detection: how high can we fly and still see the target? The rescue dummy is 1.8 m tall, 0.5 m wide. As altitude increases, the target shrinks in the camera frame. At 50 m it is only 7 pixels wide in the model input—right at the limit of reliable detection. At 70 m it disappears entirely. The hard ceiling is roughly 63 m.

2. Add a safety margin. Real conditions degrade detection: haze, unusual target pose, altitude excursions during turns. A 30% margin below the ceiling gives us **35 m operating altitude**, where the target is 36 pixels tall and 10 pixels wide—comfortably above the minimum.

3. At 35 m, how fast can we fly? The camera sees a 24 m strip of ground along the flight direction. At 4.8 frames per second (benchmarked on the Pi 5), the target must stay in view for at least 10 frames to ensure reliable multi-frame detection. That caps speed at 11.5 m/s. We chose **8 m/s**—giving 14 frames per flyover and a cumulative detection probability above 99.97%.

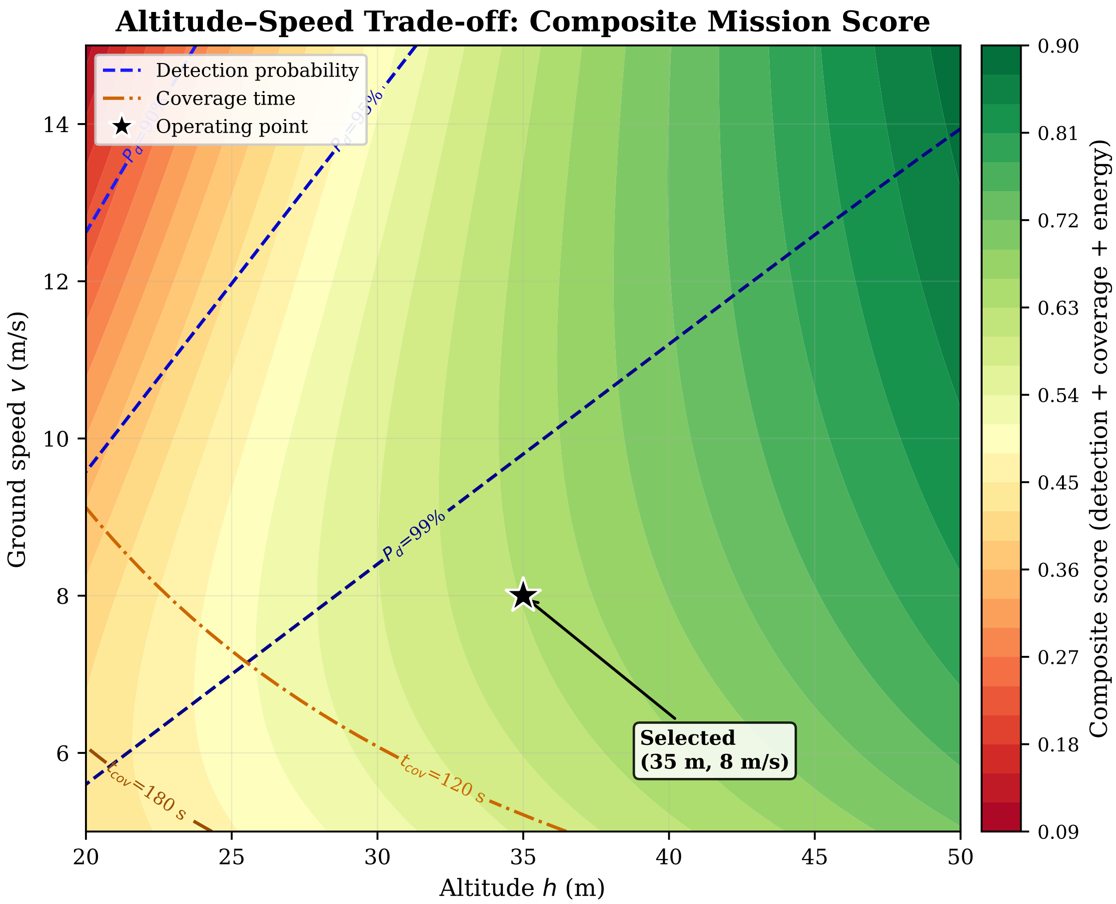

Figure 3: Altitude–speed trade-off space. Colour encodes the composite score (40% detection + 35% coverage + 25% energy). Blue contours: detection probability. Orange contours: coverage time. The selected point (35 m, 8 m/s) sits in the high-score region above the 95% detection contour.

4. What scan angle minimises energy? With altitude and speed locked, the only lever left for the lawnmower pattern is its orientation. We swept 36 angles and found that aligning with the polygon's longest edge (**70°**) cuts U-turns from 9 to 5 and halves the energy compared to the worst orientation (Figure 4).

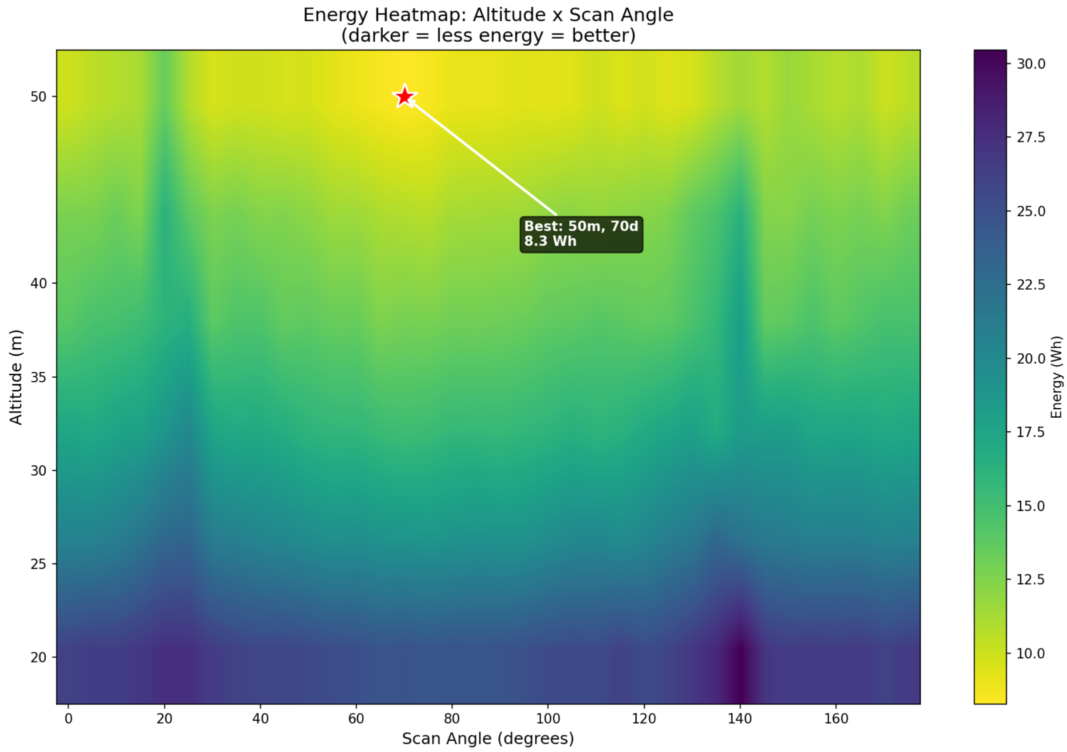

Figure 4: Energy heatmap across 216 altitude–angle configurations. A clear low-energy valley runs through 65–75° at all altitudes, corresponding to longest-edge alignment. The red star marks the global energy minimum (50 m, 70°). The selected point (35 m, 70°) trades 4.4 Wh of energy for a 44% increase in target pixel extent.

5. How much NFZ margin do we need? At 35 m the camera footprint extends 16 m from directly below the drone. Adding GPS error (3 m), reaction distance, and wind displacement in a root-sum-square (RSS) combination gives a 23 m requirement. Our **30 m buffer** provides a 2.2× safety factor. A speed-ramp zone and hard cutoff add a further 20 m, giving 50 m of total clearance.

6. Is 96% coverage acceptable? The buffer trims 4% of the polygon along the SSSI edge. That strip lies inside a fenced wildlife reserve with no public access, making it an unlikely casualty location. **96% coverage was accepted** as a deliberate trade-off against NFZ safety.

4 What We Evaluated

We did not rely on intuition. A physics-based simulation evaluated **216 configurations** (6 altitudes × 36 scan angles) on the real survey polygon, modelling energy consumption, detection probability, and NFZ penalties for every combination.

Table 2: Top three configurations from the 216-point sweep, ranked by composite score (40% detection + 35% coverage + 25% energy).

#	Alt	Angle	Speed	P_d	Energy	Score	Why not?
1	**35 m**	**70°**	**8 m/s**	**99.97%**	**12.6 Wh**	**0.87**	**Selected — best compromise**
2	30 m	65°	7.3 m/s	99.99%	14.8 Wh	0.85	17% more energy for 0.02% more detection
3	40 m	75°	8.7 m/s	99.94%	10.1 Wh	0.84	Target width drops to 9 px — thin margin

Configuration #1 wins because it sits at the "sweet spot" of the trade-off frontier (Figure 8): moving in either direction trades a large penalty in one objective for a negligible gain in another. Compared to #2, it saves 17% energy while sacrificing an unobservable 0.02 percentage points of detection. Compared to #3, it provides 3× more detection margin (10 vs 7 pixels of target width) for only 2.5 Wh of extra energy.

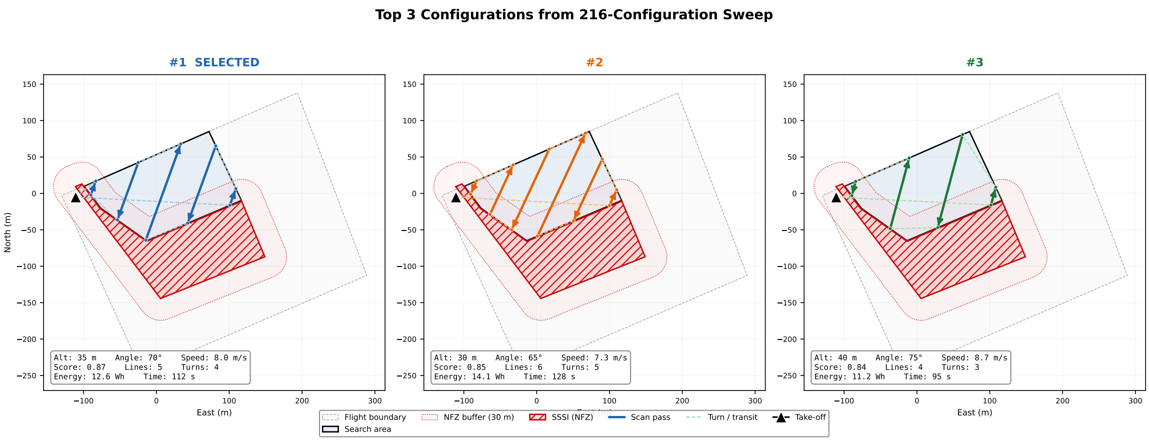

Figure 5: Top three lawnmower configurations overlaid on the survey polygon. Left: #1 (35 m, 70°, selected). Centre: #2 (30 m, 65°). Right: #3 (40 m, 75°). The SSSI no-fly zone (red) and 20 m buffer (dashed) are shown.

5 The Result

Table 3: Final operating configuration and predicted performance.

Parameter	**Value**	**Rationale**
Search altitude	35 m	30% margin below detection ceiling
Search speed	8 m/s	14 frames per target (10 required)
Scan angle	70°	Longest-edge alignment, fewest U-turns
Swath overlap	20%	Covers 4.6 m lane error with 1.8 m spare
NFZ buffer	30 m	2.2× safety factor over RSS margin
Detection probability	>99.97%	14 frames × 95% per-frame
Coverage	96%	4% gap in fenced reserve
Search time	112 s	6 scan lines + 5 U-turns
Search energy	12.6 Wh	10.9% of battery (14 min remaining)

Every parameter traces back to a physical measurement or regulatory constraint through an unbroken chain: **target size → altitude → footprint → speed → scan angle → NFZ margins → overlap**. No value was assumed or rounded for convenience.

6 Sensitivity and Robustness

The tornado sensitivity analysis (Figure 6) confirms that altitude is the most influential parameter—a 5 m change has nearly double the score impact of a 2 m/s speed change—while the scan angle has a flat optimum, meaning the pilot does not need precise compass alignment.

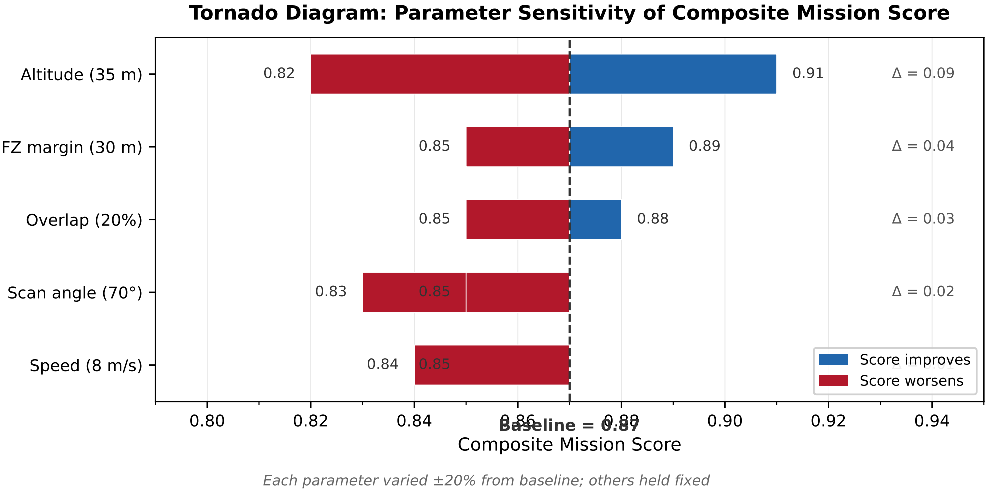

Figure 6: Tornado chart of parameter sensitivities. Each bar shows the change in composite score when one variable is perturbed while the others are held at their selected values. Altitude dominates: a ± 5 m change has nearly double the effect of a ± 2 m/s speed change.

The sensitivity spider plot (Figure 7) shows the overall sensitivity footprint: a compact polygon indicating that the selected operating point is robust to moderate perturbations in any single variable. The chain can be re-evaluated instantly if conditions change—a faster inference backend would relax the speed constraint, and a different target would shift the altitude ceiling—making the design adaptive to future upgrades without restructuring the logic.

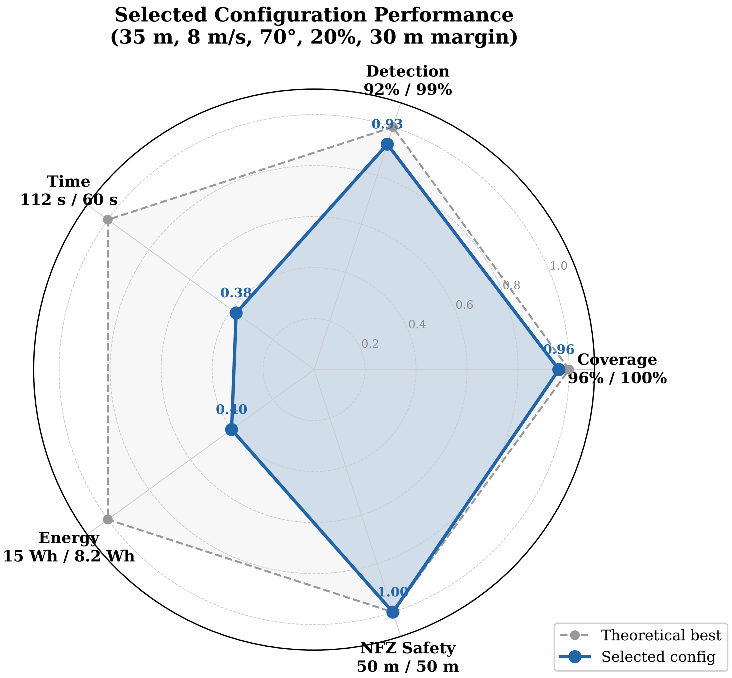

Figure 7: Spider plot of normalised sensitivities. Polygon area indicates the overall sensitivity footprint. A compact polygon means the design is robust to perturbations.

7 Pareto Optimality

The selected configuration sits at the knee of the Pareto frontier (Figure 8). Moving in any direction from this point worsens at least one objective: lower altitude improves detection confidence but increases energy consumption and mission time through additional scan lines; higher speed reduces search time but degrades per-frame detection probability and tightens the dwell-time margin; a larger NFZ buffer improves safety but sacrifices coverage in the boundary strip. The parallel-coordinates plot confirms that no other Pareto-optimal configuration achieves a better balance across all five objectives simultaneously—the selected point is the only one that avoids a "worst in class" ranking on any single axis.

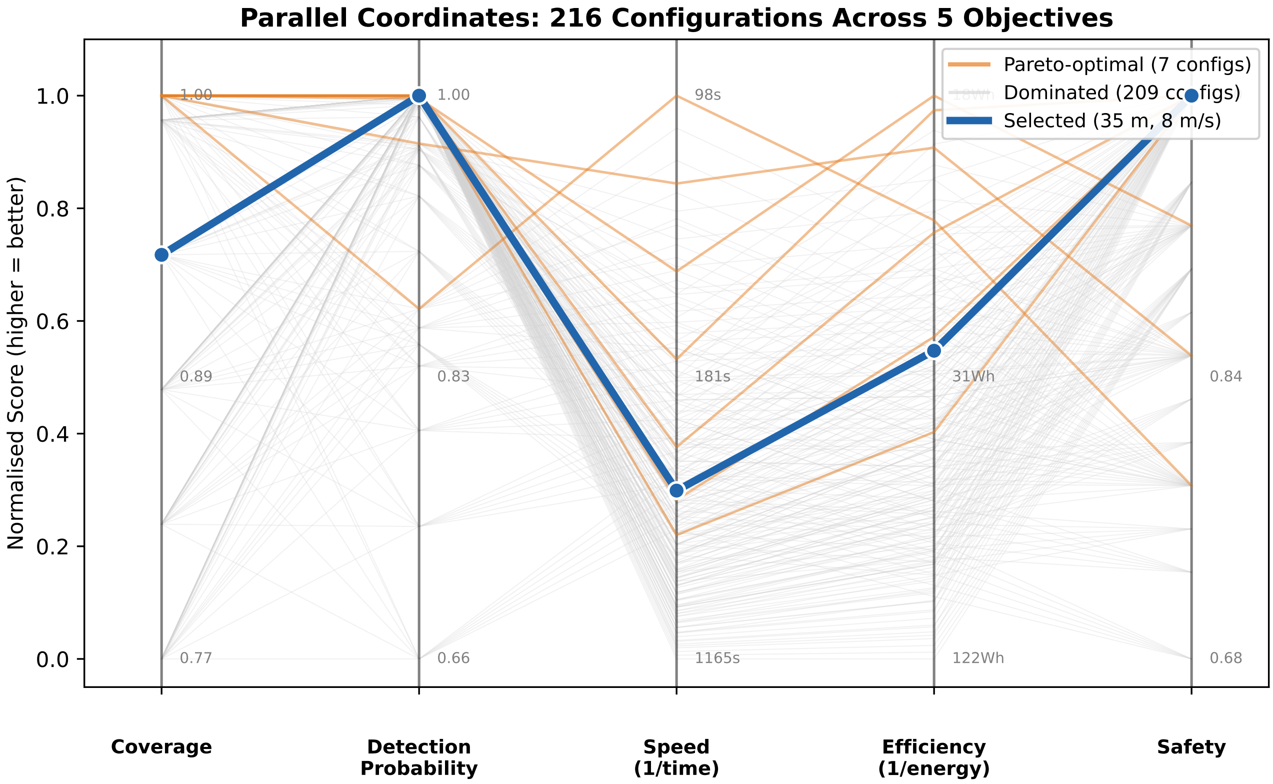

Figure 8: Parallel coordinates plot showing all five objectives simultaneously. Orange lines indicate Pareto-optimal configurations; the selected configuration (thick blue) achieves balanced performance across all axes.

8 Sequential Measurement Chain

A common pitfall in multi-objective design is attempting to optimise everything simultaneously. We avoided this by treating the design as a *sequential measurement chain*: each variable was characterised experimentally and locked before the next was addressed.

Step 1: Train a reliable vision model. The YOLOv8n model was retrained on a mixed dataset of 300 synthetic images, 16 manually labelled real flight frames, and 50 hard-negative backgrounds at the camera's native 1456×1088 resolution. Validation yielded mAP50 = 0.995. Only after confirming that the detector itself was not the bottleneck did we proceed to ask *how high* and *how fast* it could operate.

Step 2: Map the detection envelope. With a validated model, preliminary testing determined the maximum altitude at which a person-sized target remained detectable under varying lighting.

Table 4: Detection envelope: maximum reliable detection altitude under different lighting conditions.

Condition	Max detection altitude	Confidence at 35 m
Bright sun (clear day)	∼63 m	0.94
Overcast	∼52 m	0.91
Dusk / low light	∼38 m	0.85

Table 5: Speed envelope at 35 m altitude under different lighting.

Condition	Max detectable speed	Chosen speed (15% margin)
Bright sun	12 m/s	10 m/s
Overcast	10 m/s	8.5 m/s
Dusk	7 m/s	6 m/s

Step 3: Determine speed limits at the chosen altitude. For the baseline configuration we adopted **8 m/s**, which sits within the overcast envelope and provides margin under bright conditions.

Step 4: Energy-efficient path search. Only after altitude and speed were fixed did we sweep scan angle, overlap, and NFZ buffer—the 216-configuration evaluation described in Section 4.

9 Non-Quantifiable Design Choices

Not every decision admits numerical optimisation. Three choices were made on engineering judgement:

- **Lawnmower vs. spiral pattern.** A spiral can be marginally more energy-efficient for convex polygons, but a lawnmower provides a *coverage guarantee*: every point in the polygon lies within a scan lane of known width. For a safety-critical search, guaranteed coverage outweighs marginal efficiency.
- **Operator-in-the-loop verification.** Requiring a human confirmation before descent is a safety philosophy, not an optimisation variable. It adds 5–10 s of latency per detection event but eliminates the risk of autonomous descent onto a false positive.
- **Single-class detector.** Training on a single "casualty" class rather than a multi-class taxonomy simplifies the model and reduces false-positive modes. This is a deliberate simplification for first-flight reliability.

These choices reflect constraints that sit outside the quantitative model—regulatory requirements, operational risk tolerance, and project timeline.